



School of Computer Sciences
CMT221 Database Organization and Design

PROJECT GUIDELINES

I. Overview

This document provides information and guidelines on how to prepare and implement the project for the subject of CMT221: Database Organization and Design.

II. Project Objectives

The project objectives are as follows:

- To guide students in developing a database application by practicing the knowledge and theories learned in class.
- To inculcate good database design practice.
- To inculcate collaboration and documentation using a cloud-based platform.
- To familiarize with one of the Database Management Systems (DBMS).

III. Project Groups Formation

You are to work in a group of **four** members. No individual project is allowed.

All the members in a group must indicate the group formation on the spreadsheet given by the lecturers. Once the group is formed, one of the group members will be selected as the group leader.

Update the spreadsheet indicating **group number, full name of all group members and matric number, project leader, project title and SDG chosen.**

IV. Selection of Project Title

You are required to design and develop a database application for small and medium-sized enterprises (SMEs). Below are examples of general titles you can consider. You are not limited to selecting titles from the list below.

General Titles: *Logistics Database, Sales Database, Products Distribution Database, Maintenance Database, Warehouse Database, Supply Chain Database, Library Database, Membership Database, Employee Management Database, Event Management Database, Financial Database, Bill of Materials (BOM) Database, Inventory and Asset Management Database, etc.*

Please ensure that the title and your project can be mapped to one of the SDGs.

Every group must propose one title. Once your group has decided on your project title, please update your group post on the spreadsheet to include the project title and a short description of the project. Be as specific as possible when you are posting your project title. For example, the project title “University Book Library Database” instead of merely “Library Database” will give us a better idea of what your group is planning to work on. Everyone in the class should be able to view all the project titles posted on the spreadsheet. No title should be the same, meaning no more than one group can work on the exact same title. Please check the spreadsheet before you post your group title to make sure that your project does not coincide exactly with another group that has already posted on the spreadsheet.

The information about the project group and titles must be submitted by **Week 02**. The approval of the project title will be based on a first-come-first-serve basis and will be posted on the comments section by the instructors on the spreadsheet.

Based on the selected title, each member of a group must be responsible for at least one module in the project. For each module, the following requirements must be fulfilled:

- Each member must demonstrate and develop a database design which involves a **multiple-table handling** with relationship of either **one-to-many** or **many-to-many**. ✓
- Each module must **have at least three entities**. ✓
- Each member must produce a **workable sub-system with multiple forms** which are able to perform common database functions such as **adding, viewing, updating and deleting records**.
- Each member must be able to produce a **report (tabular or chart)** from the database (not the written project report).

All the modules must be integrated as a complete system. Each group has to demonstrate a database application which consists of at least four modules at the end of this semester.

V. Project Implementation

The project must be developed using the **Oracle APEX**. Oracle is used to develop the backend database while the front-end application (forms) must be able to connect to your Oracle database.

VI. Deliverables and Important Dates

The project deliverables and important dates are as follows (see Table 1):

Deliverables	WEEK	Due dates	Report Submission
Group formation and project title	2	25 October 2024, Friday	Shared Spreadsheet
Project proposal	3	1 November 2024, Friday	Softcopy
Report 1: System Design	7	1 December 2024, Sunday	Softcopy
Report 2: System Implementation	14	17 January 2025, Friday	Softcopy
System demo/presentation	15	20-24 January 2025	Slides – softcopy Demo - Webex

Table 1: Project deliverables and important dates.

Details on group formation and project title selection have been explained in Section IV. For the guidelines on how to prepare the project proposal, Report 1 and Report 2 (see Section VII).

Submission procedure: All the reports (i.e. project proposal, system design and system implementation) must be submitted by the mentioned due dates listed in Table 1. The form of the report to be submitted is also shown in Table 1. Submit a soft copy of your report in the assignment dropbox on eLearn@USM. You must submit your report before 23:59 on the deadline dates. **Late reports** without prior approval **will be penalised**.

Every group is expected to conduct a system demo at the end of the semester. Each member must participate in the system demo and explain the module(s) done by him/her. Details on the system demo (e.g. procedures, time, venue, etc.) will be made available towards the end of the semester.

VII. Documentation Format and Guidelines

This section explains the documentation format and guidelines on how to prepare documentation for the proposal, system design and system implementation.

Report Format:

All reports must comply with the following format:

- Font: Times New Roman.
- Font size: 11.
- 1.5 lines spacing.
- Content should be aligned to both left and right margins (i.e. justify).
- Page size: A4 (8.27" x 11.69").
- All pages must be properly numbered.
- For all reports (project proposal, system design and system implementation), a cover page must be included. A report with no cover page will not be marked. Refer to Section X for a sample cover page.
- All figures and tables (if any) should be labelled so that cross-referencing can be done in text.
- A table of contents (TOC) must be included (system design and system implementation only).
- A proper references and bibliography must be included in the report (if any).

Project Proposal:

The project proposal should contain (but not limited to) the following:

1. A project title.
2. A list of team members (name, matric number, USM email, role, major/minor)
3. An executive summary (overall project summary including
 - a. A brief description of the organisation you are developing the database system for
 - b. The current system and its problems
 - c. Project scope.
4. Please highlight the SDG related to the project.
5. List of modules which you intend to implement as part of the proposed project.
 - a. For each of these modules, a brief description is expected.
 - b. Indicate the module(s) assigned to each group member.

REPORT 1

System Design:

The system design should focus on the conceptual design of the proposed database. The conceptual design is performed without considering a specific DBMS and hardware (DBMS and hardware independent). The report **must not exceed 35 pages** and should contain (but not limited to) the following:

- a. Cover page.
1. Introduction **(5 marks)**.
A brief description of the organisation you are developing the database system for, the current system and its problems. Also, the motivation of the proposed project and how the proposed database system will benefit the organization must be included. Describe each module.
2. User Requirements **(10 marks)**.
Define a set of requirements describing what users can do with the database system (e.g., what data needs to be stored, what activities users must be able to perform, what features to include in the scope, etc.). Describe in detail the requirements for each module.
3. Business rules **(35 marks)**.
For each identified module stated in your project proposal, a set of business rules must be clearly stated and explained. Business rules are derived from the requirements.
4. Entity relationship modelling **(45 marks)**.
For the entity relationship diagram (ERD), please clearly state which notation (e.g. Chen notation, Crow's Foot notation or UML notation) is adapted in your report and be consistent in all ERDs.
A discussion on the ERD must be included.
5. Overall format and documentation (with proper references) **(5 marks)**.

REPORT 2

System Implementation:

The final report's **main section must not exceed 40 pages** and should contain (but not be limited to) the following: *(Some tables and diagrams can be included in the appendix (page 40 and above))*

1. Cover page.
2. All items explained in the system design report (introduction, requirements, business rules and entity relationship modelling in particular) with appropriate amendments and corrections. These amendments and corrections must be highlighted. **(10 marks)**
3. Normalization **(15 marks)**- Specify the highest normal form all modules should achieve and justify why the normal form is selected. Explain if each table has achieved the selected normal form. If a table is not in the desired normal form, show the normalization steps.
4. Data dictionary **(5 marks)**
5. Database implementation **(20 marks)**- Internal model representation via SQL commands is expected including the use of triggers and stored procedures for data manipulations.
6. Front-end system design and implementation. **(30 marks)**
 - a. Graphical user interface design (design standard and notation, screenshots, etc.).
 - b. A detailed guideline on how you design the front-end system (forms) with proper reasoning is expected.
 - c. Report designs.
 - d. User interface storyboard or flow diagram (for each module).
7. Project problems and pitfalls **(10 marks)**.
8. Conclusions/Recommendations/Future Work **(5 marks)**.
9. Overall format and documentation (with proper references) **(5 marks)**.

DEMO/Presentation

- I. Please prepare the slides and the presentation done should be in 10 minutes. 5 minutes are allocated for Questions and Answering.
- II. Please stand by 5 minutes before the presentation, and make sure all members are present and ready. Each member must present some parts of the presentation.
- III. The contents of your presentation should cover at least:

- 1. Overall Content of the slides (5%)**

- 2. Demo of system design& implementation (75%):**

- a. Tables & relationships (20%)*
- b. Forms and Reports (20%)*
- c. Data manipulations (25%)*
- d. System completeness (10%)*

- 3. Clear and confident delivery of ideas (10%)**

- 4. Understand and respond to questions (10%)**

Please refer to the full Rubrics.

PLEASE PREPARE A PRESENTATION SLIDE – to be shown during the Presentation as the introduction to your project

Some tips on the slide's content:

- The introduction with objectives
- The modules – who oversees each module
- The ERD diagram of each module and the overall ERD

Then for the **DEMO** – open **ORACLE APEX** and **show the implementation**

VIII. Project Evaluation Rubrics

The project contributes 25% of your overall grade. The marks distribution of the project is as follows:

Assignment (Project)	25%
• Group Formation & Proposal	
• Report 1	• 10%
• Report 2	• 10%
• Presentation/ Demo	• 5%

The system design, system implementation and system demo will be evaluated according to their corresponding grading rubrics which are downloadable from the eLearn@USM. The report will be evaluated individually based on your selected module.

IX. Appendix – Sample Cover Page

See next page.



SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA

CMT221 : Database Organization and Design
Semester II, Academic Session : 2024/2025

Project Proposal / System Design / Final Report

[Group Number]

[LECTURER's Name – Assigned to the group]

[Project Title]

Names	Matric No	USM Email Addressess	Module
Rashid Sidque (Leader)	10001	rash@student.usm.my	ABC
Lim Chong Wei	10002	lcw@student.usm.my	DEF
Thinaah Murali	10003	thinaah@student.usm.my	GHI
Watson Nyambe	10004	watz@student.usm.my	JKL

Date of submission
1 December 2024

RUBRICS

Report 1

Report elements	Does not meet standard (1 - 3 points)	Meets standard (4 - 7 points)	Exceeds standard (8 - 10 points)
Introduction (5%)	The introductory section is vague.	The introductory section is clearly stated with a focus.	The introductory section is clearly stated, has a sharp, distinct focus and enhances the impact of the report.
Requirements (10%)	Requirements are vague or poorly defined.	Requirements are adequately defined but not in-depth and detailed.	Requirements are complete, in-depth, and detailed.
Business rules/Process flow (35%)	The presented business rules are not complete, vague and imprecise.	The presented business rules are complete but not in-depth and detailed.	The presented business rules are complete, in-depth, and detailed.
	The identified business rules are ambiguous.	The identified business rules are somehow ambiguous.	The identified business rules are not ambiguous.
Entity-relationship modeling (45%)	The selected entities and relationships in an ERD of a particular module are modelled imprecisely. The student demonstrates minimal ability to present design concepts in a basic ERD.	The selected entities and relationships in an ERD of a particular module are modelled correctly but some missing information in terms of relationship strength, connectivity, cardinalities and keys are identified.	The selected entities and relationships in an ERD of a particular module are completely modelled with proper relationship strength, connectivity, cardinalities and all keys appropriately documented.
	The choice of ERD notation is clearly stated.	The choice of ERD notation is clearly stated.	The choice of ERD notation is clearly stated.
	No discussion based on a particular module is included.	A discussion based on a particular module is included but not in-depth and detailed.	An in-depth and detailed discussion based on a particular module is included.
Overall documentation (5%)	The requirements listed in the project guidelines are not met.	The requirements listed in the project guidelines are met but with some exceptions.	All the requirements listed in the project guidelines are met.
	Many spelling and grammatical mistakes are identified (more than 20).	Some spelling and grammatical mistakes are identified (between 10 to 20).	Minimal spelling and grammatical mistakes are identified (less than 10).

Report 2

Report elements	Does not meet standard (1 - 3 points)	Meets standard (4 - 7 points)	Exceeds standard (8 - 10 points)
First report items (executive summary, intro, business rules and ER model (10%))	The mentioned items are not correctly amended in accordance to the supervisor's comments.	The mentioned items are correctly amended in accordance to the supervisor's comments but mistakes are still identified.	The mentioned items are correctly amended in accordance to the supervisor's comments.
Normalization (15%)	The design reflects a student owns some basic skill in table design and normalization.	The design reflects tables in 3NF.	The design reflects tables in 3NF or rationale to choose other normal form.
Data dictionary (5%)	Inconsistent data type selection for fields (more than 10). No naming convention is proposed and adhered to throughout the design.	Data types for fields show an understanding of the problem and the type of data to be stored. However, some inconsistencies are identified (between 5 to 10). Naming convention is proposed but it is not strictly adhered to throughout the design.	Data types for fields show a thorough understanding of the problem and the type of data to be stored. The naming convention is selected and adhered to throughout the design.
Database implementation (20%)	The SQL script (DDL) is appropriate but contains some syntax errors.	The SQL script (DDL) is appropriate for selected tables and is syntactically correct.	The SQL script (DDL) is complete and ready for database implementation.
Front-end system design and implementation (30%)	No GUI design standard is proposed. Important screenshots are not explained in accordance to the proposed standard.	A GUI design standard is proposed but they are not explained appropriately with some screenshots.	A complete and detailed GUI design standard is proposed. Important screenshots are explained in depth in accordance to the proposed standard.
Project problems and pitfalls (10%)	The student is not able to perform reflection based on the knowledge and experience gained throughout the project implementation.	The student can perform reflection based on the knowledge and experience gained throughout the project implementation.	The student can perform reflection based on the knowledge and experience gained throughout the project implementation. The student also shows how these problems have been overcome.
Conclusions / Recommendations / Future Work (5%)	The conclusion section is superficial. No recommendations or future works are proposed.	The concluding paragraph follows and summarizes the report discussion, and draws a conclusion.	The concluding section summarizes and draws a clear, effective conclusion and enhances the impact of the report. Recommendations and future works are highlighted.
Overall documentation (5%)	The requirements listed in the project guidelines are not met.	The requirements listed in the project guidelines are met but with some exceptions.	All the requirements listed in the project guidelines are met.
	Many spelling and grammatical mistakes are identified (more than 20).	Some spelling and grammatical mistakes are identified (between 10 to 20).	Minimal spelling and grammatical mistakes are identified (less than 10).

Presentation

Criteria	%	Very good	Good	Fair	Weak	Very weak
		9-10	7-8	5-6	3-4	1-2
Overall Content of the slides (5%)	5	Slides with relevant information. Topic is discussed clearly and adequate points are made.	Slides with relevant information . Topic is somewhat discussed but more information is required.	Slides with relevant information. Topic is slightly discussed but more material is needed.	Less slides that contain relevant information. Topic is addressed but not discussed or elaborated on.	Slides do not contain relevant information. Topic is not addressed and not discussed or elaborated on.
Demo of system design& implementation (75%):						
Tables & relationships (20%) Multiple-table (at least two) is implemented. Primary key(s) is/are defined in each table. Proper naming convention is applied in accordance to the data dictionary. An appropriate number of fields are defined in each table. Each table is inserted with at least 20 meaningful records. All relationships between tables are specified (one-to-many or many-to-many). Referential integrity is enforced for all relationships. Foreign key(s) is/are used appropriately. show the normalization steps.	20	The student has created an EXCELLENT tables & relationship design which includes ALL or above 90% of the mentioned features.	The student has created a GOOD tables & relationship design which includes 70 - 90% of the mentioned features.	The student has created an ACCEPTABLE tables & relationships design which includes some of the mentioned features (50%-70%).	The student has created an WEAK tables & relationships design which includes some of the mentioned features (30%-50%).	The student has created an UNACCEPTABLE tables & relationships design which includes some of the mentioned features (less than30%).
Forms and Reports (20%) Validation rules , input masks and lookup fields are appropriately used Forms are included for users to enter data into all tables. Design of the forms is logical, customized and aesthetically pleasing. Meaningful reports are created based on a particular module. All reports in the database application are consistent in terms of look and feel. Reports are sorted and summary fields are used when appropriate.	20	The student has created an EXCELLENT Data manipulations which includes ALL or above 90% of the mentioned features.	The student has created a GOOD Data manipulation s which includes 70 - 90% of the mentioned features.	The student has created an ACCEPTABLE Data manipulations which includes some of the mentioned features (50%-70%).	The student has created an WEAK Data manipulations which includes some of the mentioned features (30%-50%).	The student has created an UNACCEPTABLE Data manipulations which includes some of the mentioned features (less than30%).
Data manipulations (25%) Search record function is implemented in all forms (queries). A new record can be added. An existing record can be updated or deleted. Stored procedures and triggers are created and used to manipulating data.	25	The student has created an EXCELLENT Data manipulations which includes ALL or above 90% of the mentioned features.	The student has created a GOOD Data manipulation s which includes 70 - 90% of the mentioned features.	The student has created an ACCEPTABLE Data manipulations which includes some of the mentioned features (50%-70%).	The student has created an WEAK Data manipulations which includes some of the mentioned features (30%-50%).	The student has created an UNACCEPTABLE Data manipulations which includes some of the mentioned features (less than30%).

System completeness (10%) A switchboard form is implemented. The proposed GUI standard is adhered to. All modules are integrated seamlessly. Tables, forms and reports are fully functional and integrated consistently.	10	The system is fully complete with includes ALL or above 90% of the mentioned features.	The system is partially complete which includes 70-90% of the mentioned features.	The system is partially complete but includes some of the mentioned features (less than 50%-70%).	The system is incomplete but includes some of the mentioned features (less than 30%-50%).	The system is incomplete but includes some of the mentioned features (less than 30%).
Clear and confident delivery of ideas (10%)	10	Able to deliver ideas with great clarity and confidence	Able to deliver ideas clearly and confidently	Able to deliver ideas fairly clearly and confidently, thus require minor improvements	Able to deliver ideas with limited confidence and require further improvements	Not able to deliver ideas clearly and confidently and require major improvements
Understand and respond to questions (10%)	10	Able to fully understand and respond to questions very well	Able to respond to questions well	Able to understand and answer questions satisfactorily	Able to understand and answer questions but not able to accurately answer the question	Not able to understand and respond to a question